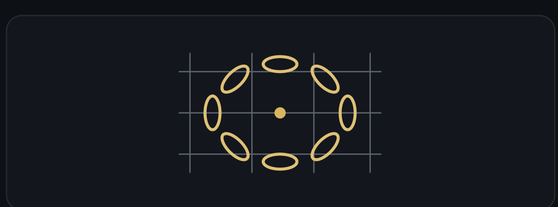
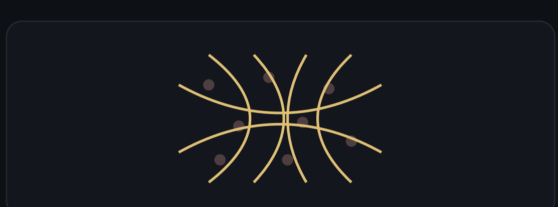
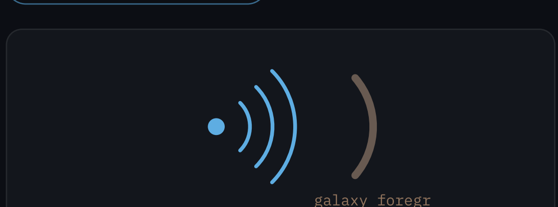
1 · WHAT WE LOOK AT Light from matter Gravity bending light Early universe, direct Spacetime itself	2 · WHAT IT TRACES	3 · WHEN IN COSMIC HISTORY 0 1 2 10 100 1100 · z →	4 · WHAT IT TELLS US	5 · MAIN DIFFICULTY
01 Galaxy Clustering GEOMETRY + GROWTH 	BIASED TRACER Galaxy positions — a biased echo of the invisible dark-matter web	$z = 0 - 2$ last ~10 billion years	STANDARD RULER Expansion & growth rate — plus the frozen BAO sound-wave ruler	GALAXY BIAS Galaxies aren't dark matter — the link between them must be modelled
02 Weak Gravitational Lensing GROWTH 	ALL MASS All intervening mass — dark and visible alike, with no assumptions	$z \approx 0.5 - 1.5$ sources · structure at z ~ 0.5	MATTER & CLUMPING How much matter there is, and how clumpy it is	1% SIGNAL A ~1% distortion, hidden in galaxy noise — needs 100 million galaxies
03 Cosmic Microwave Background GEOMETRY · INITIAL CONDITIONS 	SOUND WAVES Sound waves in the infant plasma, frozen as it turned transparent	$z = 1100$ 380,000 yr after the Big Bang snapshot	THE ANCHOR The initial conditions that calibrate every other probe	ONE SKY A single 2D snapshot, and only one sky exists — cosmic variance
04 CMB Lensing GROWTH 	LIGHT BENT All mass warping the primordial light on its 13-billion-year journey	$z \sim 2$ lensing source fixed at z = 1100	NEUTRINO MASS Growth where gravity is still simple — the best route to weighing neutrinos	RECONSTRUCTED A subtle warp that must be reconstructed statistically — never seen directly
05 The Lyman-α Forest GROWTH · SMALL SCALES 	CORE SAMPLE Intergalactic hydrogen — each quasar drills a core sample through the web	$z = 2 - 4$ deeper than any galaxy survey	SMALL SCALES The smallest scales, deepest reach — probes neutrinos & warm dark matter	GAS TEMPERATURE The uncertain gas temperature history mimics the very signal we're after
06 Gravitational Waves GEOMETRY · DISTANCE 	SPACETIME Ripples in spacetime itself — nothing to do with light or matter	$z < 1$ → 10 now → future detectors	ABSOLUTE DISTANCE Distance directly from GR, no calibration ladder — an independent H_0	NO REDSHIFT It gives distance but not redshift — that must come from elsewhere
07 21 cm Intensity Mapping GEOMETRY + GROWTH 	NEUTRAL HYDROGEN Neutral hydrogen everywhere — each frequency channel is a distance slice	$z = 0\text{–}3 \cdot 6\text{–}30$ mature universe · first stars	HUGE VOLUMES The largest volumes ever surveyed — and the only view of the dark ages	FOREGROUNDS Our galaxy shines 10,000× brighter — a whisper beside a jet engine